Masked Topology Convolutional Network for Classification and Segmentation of Remote Sensing Images

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***Abstract*— Convolutional neural networks have made significant progress in remote sensing image processing. Convolutional networks mostly model the local information of samples based on pixel features, while ignoring the topological relationship among different categories of ground objects. However, the nonlocal topology can better represent the underlying data structure of the image. In order to make up for the insufficiency of Convolutional neural networks in extracting nonlocal topological information, we propose a new masked topology convolutional network (MTC_Net). First, considering the regionality of different categories of ground objects, we use a clustering algorithm to simply cluster the pixels to extract the most obvious area mask, which can strengthen the boundary information among different ground objects. Second, we take the cluster center point as the representative point of the region and use the graph structure to represent the topological relationship, in addition use graph convolution to further extract the topological structure relationship among different categories of ground objects. Finally, we use convolution modules and multilevel feature attention modules to capture important local and global contextual semantic information and reduce category confusion. We conduct a great deal of experiments on four (two classification and two segmentation) public datasets, and the proposed MTC_Net achieves excellent classification and segmentation performance.**

***Index Terms*— Classification and segmentation, graph convolutional network (GCN), remote sensing images, topological structure.**

I. Introduction

REMOTE sensing image processing is the process of processing and analyzing images acquired from remote sensing devices such as satellites, aircraft, or drones. It can obtain various information on the surface, monitor environmental changes [1], provide disaster monitoring and emergency response [2], be used for military reconnaissance and security monitoring [3], as well as scientific research. Remote sensing image processing is of great significance in the fields of geographic information system [4], urban planning [5], environmental monitoring [6], and so on.

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Along with the development of computer technology, an increasing number of methods are applied to the processing of remote sensing images. Fu et al. [7] proposed a method based on image scattering structure features for template matching was proposed to achieve aircraft recognition. Li et al. [8] combined the sparse autoencoder and the K_means clustering method to propose an unsupervised domain adaptation method. The source domain and the target domain are adaptively mapped by learning a shared feature space to achieve the result of classification. In [9], a mutual boosting attention transformer (MBATrans) approach to capture cross-domain dependencies of semantic feature representations was proposed, which not only aligns cross-domain features but also enhances general functions. Ma et al. [10] proposed a hyperspectral image classification method based on unsupervised spectral motion feature learning. This method converts the spectral changes in hyperspectral images into spatial changes by learning the characteristics of spectral changes, but it may not be effective for hyperspectral images with small spectral changes.

Existing supervised learning methods work best with deep neural networks [11], [12], [13], [14], [15], [16], [17]. Cao et al. [11] proposed a lightweight multiscale neural architecture search method for hyperspectral image classification, using a novel spectral-spatial attention mechanism that can improve the classification performance meanwhile without increasing computational complexity. Jha et al. [14] proposed a fusion network based on global attention, using a new self-attention and cross-attention learning technology to perform classification verification on remote sensing datasets of various modalities, and obtained excellent results. Gao et al. [15] proposed the depthwise feature interaction network for wetland classification, in this way, meaningful complementary information is emphasized for classification, and excellent performance is achieved in multisource remote sensing image classification tasks. Gu et al. [17] proposed a novel network that combines CNN and Transformer mechanisms. However, the existing problem is that the calculation of the Transformer mechanism is complex, it cannot handle large images, so the image needs to be cropped into small images, which will lead to the loss of global information on the image.

The inherent topological relationship between different categories in the image is not limited by coordinates and maintains its original properties even when deformed. This invariant

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feature is beneficial for improving classification performance. Xu et al. [18] proposed a spatial-aware graph relation network (SGRN) to adaptively discover and integrate key semantic and spatial relationships for reasoning about each object, used for learning an interactive sparse graph structure to encode relevant contextual regions. Gupta et al. [19] proposed a new topological interaction module that encodes topological interactions into deep neural networks, which showed excellent performance in multiclass medical image segmentation tasks. These methods usually transform topological data into simple feature representations through some data processing methods, but ignore the spatial dependency of information on certain nodes.

After the above analysis, the traditional methods such as clustering and template matching usually have excellent interpretability for features, but they rely on hand-designed parameters and rules, which limits their feature description capabilities. Deep Neural Network can effectively extract context information to improve segmentation\classification accuracy. However, most methods model the local information of the sample based on pixel features, while ignoring the topological relationship among different categories of ground objects, but nonlocal topology can better represent the underlying data structure of the image. How to extract more effective information based on the intrinsic characteristics of remote sensing images and improve the accuracy of downstream tasks.

In order to solve the above issues, further fill the insufficiency of convolutional neural networks in extracting nonlocal topological information, we propose a novel masked topology convolutional network (MTC_Net), which enables efficient segmentation and classification of remote sensing images. The main contributions are summarized as follows.

1) A new MTC_Net is proposed, which utilizes a clustering algorithm to extract area masks and uses a graph structure to represent topological relationships, as well as a multilevel feature attention fusion (MLAF) module, which achieves superior classification and segmentation performance.
2) We proposed a mask generation algorithm (MGA) that can perform simple clustering on pixels, compared to the traditional SLIC algorithm, extract the most obvious area mask, and strengthen the boundary information among different ground objects.
3) We proposed GCN-based topological information extraction model that can further extract the topological information between different categories of ground objects, which is beneficial to enhance the Interrelationships between ground objects. This is a feature that existing algorithms have not extracted for remote sensing images.
4) The MLAF module aims to capture local and global important contextual semantic information, reduce category confusion, and conduct a large number of experiments on four public datasets, achieving good classification and segmentation performance.

The results of ablation experiments can prove the effectiveness of MTC_Net. The rest of this article is organized as follows. In Section II, we briefly introduce related work. Section III mainly explains the theoretical details of MTC_Net. In Section IV, the relevant experimental results of MTC_Net are displayed and analyzed. Finally, Section V is the conclusion.

II. Related Works

A. Clustering for Remote Sensing Images

Xie et al. [20] proposed a clustered feature-constrained multiscale attention network (CFANet) for shadow extraction in remote sensing images. The network combines a clustering algorithm and a multiscale attention mechanism to improve the accuracy and robustness of shadow extraction. Ding et al. [21] proposed a hyperspectral image classification method based on clustering heuristic active learning, combining clustering algorithm and active learning, and using a clustering algorithm to improve the efficiency and accuracy of sample selection. Pal et al. [22] proposed a variable-length multiobjective genetic clustering method for multiclass change detection, which combines the variable-length multiobjective genetic clustering method with support vector machines to improve the accuracy and efficiency of high-resolution satellite image segmentation and change detection.

The above methods combine the clustering algorithm with other methods to improve the performance of the overall architecture, which provides a theoretical basis and performance proof for the use of clustering algorithms to generate remote sensing image mask.

B. Graph Convolutional Network (GCN)

GCNs [23] is a deep learning model based on graph-structured data, first proposed by Thomas Kipf and Max Welling in 2016. Later, along with the development of computer technology, GCNs were gradually applied to remote sensing image processing. Li et al. [24] proposed a deep learning classification method based on GCNs, which first represents remote sensing images as graph-structured data, where each pixel is represented as a node in the graph, and the edges among nodes represent the spatial relationships among pixels. Then, GCNs are used to perform convolutional operations on the graph to extract features of the image and use fully connected layers for classification. Zhu et al. [25] proposed a hyperspectral image classification method based on multiscale short-range and long-range GCNs, which can efficiently handle graph-structured data and outperform other methods in terms of classification accuracy. Fusing convolutional neural networks and GCNs is also an effective approach. Zhou et al. [26] effectively combine attention multihop graph convolutional and multiscale convolutional methods. The multihop GCN systematically extracts structural information and achieves good results on several remote sensing classification datasets. Duan et al. [27] consider the high-order structures among multiple pixels and propose a structure-preserved hyper GCN (SPHGCN). SPHGCN optimizes the distribution of convolutional features guided by high-order structures while obtaining more discriminative features from multiple pixel nodes. Experimental results on three HSI datasets demonstrate

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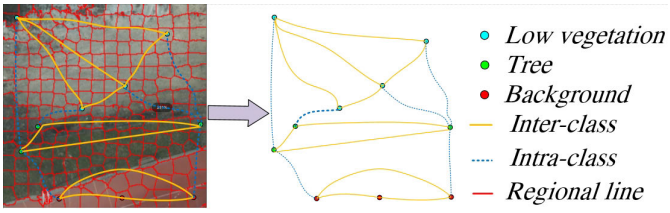


Fig. 1. Schematic of the topological relationship of regional representative points in remote sensing images.

that SPHGCN outperforms some state-of-the-art HSI classification methods.

From the analysis of experimental results, GCNs can further improve the processing efficiency of the model, which also provides a theoretical basis for MTC_Net.

C. Topology Structure of Images

The topological structure of an image refers to the spatial relationships among pixels in the image. In a 2-D image, the spatial relationships among pixels can be represented as a 2-D grid, where each pixel is represented as a node in the grid, and the edge between adjacent pixels is represented as an edge in the grid, as shown in Fig. 1. Kang et al. [28] proposed a method for aircraft classification in high-resolution synthetic aperture radar (SAR) images based on scattering topology network (ST-Net). First, the SAR image is transformed into a scattering coefficient image using wavelet transform, and then the ST-Net is used to classify the scattering coefficient image. In addition, the method also has good generalization ability and robustness to noise and variation in the input data. In [29], a cross-scene hyperspectral image classification method based on topology and semantic information transfer network was proposed, which can effectively deal with the topological structure of images and outperform other methods in terms of classification accuracy. Wang et al. [30] proposed a high-resolution satellite image urban land use classification method based on topological data analysis (TDA).

The above methods demonstrate that using the topological structure information of remote sensing images can further extract image features.

III. Methods

In this section, we elaborate on the overall framework of MTC_Net and its components, including the MGA, image topological construction, GCN, and the construction of MLAF modules and the objective function.

A. Overview of MTC_Net Framework

The classification framework of MTC_Net is shown in Fig. 2. It mainly includes five parts, namely MGA, the extraction of the image topological structure and GCN, the basic framework of the convolutional network, the MLAF module, and the construction of the objective function.

The basic framework of MTC_Net is improved and innovated on the basis of VGG16 [31]. VGG16 has a simple architecture but powerful feature extraction capabilities. Therefore, we choose VGG16 as the backbone. Compared with other deep learning algorithms, they only model the contextual semantic information, while ignoring the topological relationship among different categories of ground objects, but the nonlocal topology can better represent the underlying data structure of the image. Therefore, image topological structure information is incorporated into the VGG16 network architecture, while utilizing VGG16 to extract contextual semantic information. This structure not only guarantees the extraction of image topological structure information but also fully captures the contextual semantic information, enabling our proposed method to achieve excellent performance in both classification and segmentation tasks.

The input image is $\boldsymbol{I} \in \mathbb{R}^{H\times W\times C}$, first input $\boldsymbol{I}$ into the MGA, and the output after algorithm processing is $\boldsymbol{I_{mask}} \in Cls^{H\times W\times 1}$, $Cls \in \{1, 2, \ldots, M\}$, M is the set number of categories, and at this time $\boldsymbol{I_{mask}}$ is the area mask based on the center point pixel category, In the process of extracting contextual semantic information, we concatenate $\boldsymbol{I}$ and $\boldsymbol{I_{mask}}$ to generate a mask image $\boldsymbol{I_{in}} \in Cls^{H\times W\times(C+1)}$ and input it to the convolutional network to extract contextual semantic information.

Since the convolutional network captures only local information, in order to obtain global features of the image, we use the MLAF module to extract global information. The attention mechanism is used to obtain global information, in the meantime the multiscale feature fusion is beneficial for recognizing objects of different scales and shapes. The output of the MLAF module is then passed through a fully connected layer for classification.

In the process of extracting image topological structure information, the GCN we use can further extract topological structure information. Since the GCN itself can be classified based on topological structure, we add a linear layer to the end of the GCN for classification. To fuse image topological structure information and semantic information, after the first convolution layer of the GCN, we perform global average pooling on the features generated by the GCN and input them into the second convolution operation of the convolutional network. The resulting feature is added to the previous layer's convolution feature and used for the convolution operation of the current layer.

Additionally, after the third convolution layer of the GCN, the features generated by the GCN are again subjected to global average pooling and input into the fourth convolution operation of the convolutional network. The resulting feature is added to the previous layer's convolution feature and used for the convolution operation of the current layer. This achieves the fusion of image topological and semantic features. Finally, the predicted results generated by the GCN and the convolutional network and the true labels are input into the objective function to optimize the model parameters using the backpropagation algorithm to obtain an optimal model.

B. Mask Generation Algorithm

After the original image $\boldsymbol{I}$ is input to the MGA, the output Imask [as Fig. 3(e)] is an area mask based on the center point pixel category, the advantage of doing this is that the original image can be partitioned into blocks with similar

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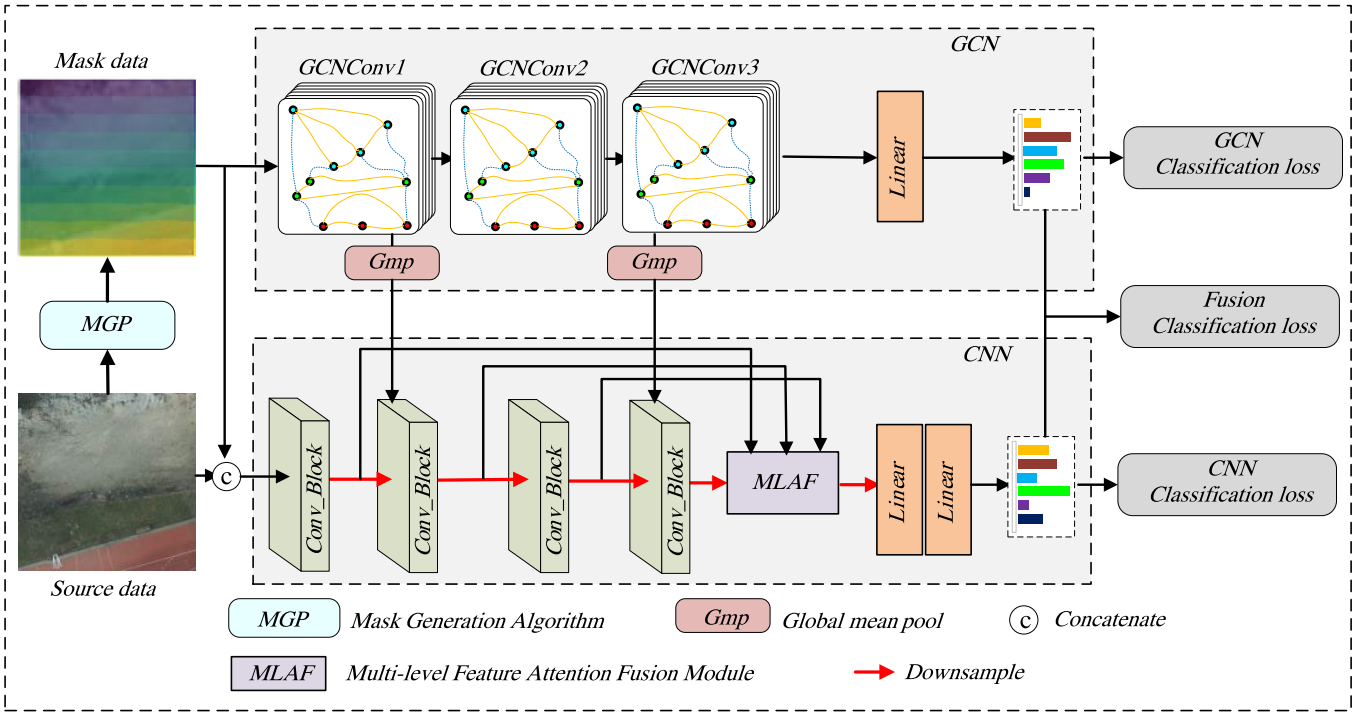


Fig. 2. Overall framework of the MTC_Net classification network. the framework consists of four main parts. On the left side, there is an image mask generated using the MGP algorithm. The upper part represents the GCN, which consists of three graph convolutional layers and one linear layer. The lower part represents the convolutional network, which consists of four convolutional blocks, one MLAF module, and two linear layers. The GCN and the convolutional network are connected through global average pooling. On the right side, there is the loss fusion module used in this framework.

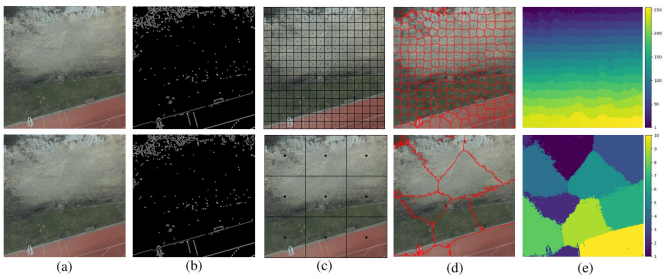


Fig. 3. Visualization of the intermediate results of the MGA algorithm, The first row shows the visualization of each stage when the number of clusters is 256 ($M = 256$). The second row shows the visualization of each stage when the number of clusters is 10 ($M = 10$). (a) Original image as I in Section III-B. (b) Image after gradient calculation of image I, corresponding to G in Section III-B. (c) Initialization of cluster center positions, and the black grid is used as a reference line to obtain the cluster centers. (d) Visualization of the cluster centers and cluster areas obtained after G has undergone ten rounds of clustering, corresponding to Dist in Section III-B. (e) Visualization of the image after replacing the pixels in each cluster area with the cluster number, corresponding to Mask in Section III-B.

colors, textures, and shapes using the mask information. At this time, the mask not only includes the object boundaries in the image but also provides key point information for extracting the topological structure of the image. Fig. 3 illustrates the visualization of some intermediate results of the MGA algorithm. First, the cluster centers are initialized as the center points of $S \times S$ squares [Fig. 3(c)]. Then, the clustering algorithm (Algorithm 1) is applied to refine the cluster center points and cluster regions, resulting in the final cluster center points and cluster regions [Fig. 3(d)]. We assign labels to the clusters in an increasing order from left to right and top to bottom. Each pixel within each cluster region (red block) is replaced with the label of the cluster center point, generating our mask image [Fig. 3(e)], which exhibits a horizontal effect in visual appearance. The reason for the horizontal topology matrix is that we construct the cluster labels in increasing order from left to right and top to bottom, representing the category information of all pixels within each cluster region. Instead of performing clustering on the overall image level (which would result in fewer clusters and larger cluster regions), we perform clustering within smaller regions. This is because when passing the input into the GCN, the input is the cluster center points. When the number of cluster categories is too small, it cannot effectively capture the topological information of the image. The MGA algorithm is described in Algorithm 1.

The size of the original image $\boldsymbol{I}$ is $W \times H$, and the pixel value of each pixel is (r, g, b). First, we perform the clustering initialization process to generate M initial cluster centers, the initial C_i is generated according to (1) and (2)

$$X = s \times d, \quad s = \left\lfloor \sqrt{\frac{W \times H}{M}} \right\rfloor, \quad d = \left[1, 2, \ldots, \left\lfloor \sqrt{M} \right\rfloor\right] \tag{1}$$

$$Y = s \times d, \quad s = \left\lfloor \sqrt{\frac{W \times H}{M}} \right\rfloor, \quad d = \left[1, 2, \ldots, \left\lfloor \sqrt{M} \right\rfloor\right] \tag{2}$$

$$C_i = [w, h] = [\langle X, Y \rangle]_i \tag{3}$$

where M is a hyperparameter, indicating the number of cluster points. s is the step size selected by the cluster center. d is

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the position multiplier. $\lfloor\rfloor$ indicates floor operation. $\boldsymbol{X}$ contains $(M)^{1/2}$ values, $\boldsymbol{Y}$ also contains $(M)^{1/2}$ values, then C contains M coordinate pairs.

The direction of the gradient is the direction in which the function changes fastest, so when there is an edge in the function, there must be a larger gradient value. On the contrary, when there are relatively smooth parts in the image, and the grayscale values change less, the corresponding gradient is also smaller. In the image processing, the modules of the gradient is simply referred to as gradient, and the image composed of image gradients is called a gradient image. For subsequent operations, we first grayscale the original image $\boldsymbol{I}$ [as Fig. 3(a)] into $\boldsymbol{g}$, and then we calculate the gradient image $\boldsymbol{G}$ [as Fig. 3(b)] of $\boldsymbol{g}$ according to the following equations:

$$\nabla f(x, y) = \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix} = \begin{bmatrix} f(x+1, y) - f(x-1, y) \\ f(x, y+1) - f(x, y-1) \end{bmatrix} \quad (4)$$

$$G = \|\nabla f(x, y)\|_2 \quad (5)$$

where $x \in \boldsymbol{X}$ and $y \in \boldsymbol{Y}$ represent the coordinates of the pixel point. $f(x, y)$ represents the grayscale value of that pixel. $\|\cdot\|_2$ represent computing the L_2 norm.

Then, fine-tune the cluster center point, project the cluster center C_i onto $\boldsymbol{G}$, and find the position with the smallest gradient in the 3×3 field of $\boldsymbol{G}$ position C_i and then replace C_i. This is done to avoid locating cluster centers on edges, reducing the effectiveness of clustering with domain pixels.

Algorithm 1 Mask Generation Algorithm

Input: Original image $\boldsymbol{I}$, maximum number of iterations T, and number of clustering centers M.
Initialization: Initialize the clustering center points C_i, gradient image $\boldsymbol{G}$, $\boldsymbol{Mask}$, and $\boldsymbol{Dist}$ matrix according to equations (1)-(5).
for $t = 1, 2, \ldots, T$ **do**
  Fine-tune C_i based on the 3×3 neighborhood of the gradient image $\boldsymbol{G}$ and clustering center C_i.
  for $i = 1, 2, \ldots, M$ **do**
    for Each pixel p within the $2s$ neighborhood of the clustering center C_i **do**
      Calculate the chromatic distance $\boldsymbol{D}$ between p and C_i based on equations (6)-(8);
      if $D < Dist(p)$ **then**
        set $mask(p) = i$
        set $Dist(p) = \boldsymbol{D}$
      end if
    end for
  end for
  Calculate the center position of the region with C_i as the center and s as the neighborhood radius, and replace it with the new C_i.
end for
Output: The $\boldsymbol{mask}$ matrix and the coordinates of clustering center points $\boldsymbol{C_i}$.

Next, initialize a $\boldsymbol{Mask}$ matrix and $\boldsymbol{Dist}$ matrix. The size of $\boldsymbol{Mask}$ and $\boldsymbol{Dist}$ is $W \times H$, and then project C_i onto the $\boldsymbol{Mask}$, and initialize the value of the area with C_i as the center and the neighborhood radius is s to i. The $\boldsymbol{Dist}$ matrix is initialized to ∞, $\boldsymbol{Mask}$ is used to save the category of each pixel, and $\boldsymbol{Dist}$ is used to save the chromatic distance between the pixel and the nearest clustering center.

For each cluster center, calculate the chromatic distance between all pixels within the 2 s neighborhood radius and the cluster center position according to the following equations. There will be overlapping areas in the adjacent cluster midpoints, that is to say, some pixels belong to the neighborhood of multiple cluster centers. At this time, we choose the one with the smallest chromatic distance as the category of the point, and update the $\boldsymbol{Mask}$ and $\boldsymbol{Dist}$ [as Fig. 3(d)] matrices. The calculation of the chromatic distance between the pixel point $P_j(x_j, y_j)$ and the cluster center $C_i(x_i, y_i)$ refers to the following equations:

$$d_c = \sqrt{\left(r_j - r_i\right)^2 + \left(g_j - g_i\right)^2 + \left(b_j - b_i\right)^2} \quad (6)$$

$$d_s = \sqrt{\left(x_j - x_i\right)^2 + \left(y_j - y_i\right)^2} \quad (7)$$

$$D = \sqrt{d_c^2 + d_s^2} \quad (8)$$

d_c and d_s, respectively represent the chromatic distance and the Euclidean distance between the pixel point P_j and the cluster center point C_i, and the $\boldsymbol{Mask}$ and $\boldsymbol{Dist}$ of each pixel point can be generated by traversing the cluster center.

For each cluster center, an area will be formed after the above steps, and then the center position of this area will be calculated and updated as the cluster center point of this area, subsequently the cluster center point will be fine-tuned through the gradient image to form a new cluster Center, so that the above steps are repeated multiple times to generate the final image $\boldsymbol{Mask}$.

C. Image Topological Construction

Since the convolutional network mainly extracts the local context semantic information of the image, it does not extract enough topological structure information on the image. However, the topological structure information of the image also contains rich long-distance feature information. Therefore, we want to use the topological structure information of the image further to improve the performance of the model. Due to the large number of pixels in the image, it is unrealistic to extract topological structure information from all pixels, but the pixels in the adjacent neighborhood contain certain similarities. Therefore, we can use key point pixels to represent the topological structure information of all pixels in the neighborhood. In this way, the calculation cost caused by the number of pixels can be greatly reduced.

Use the MGA algorithm to extract the clustering center of the image as the key point $C = \{C_1, \ldots, C_M\}$, which greatly reduces the number of pixels, then construct the topological structure of the image through these clustering center points and use the graph convolutional neural network further to extract topological structure information. First, these key points C are represented by the graph structure $G = (V, E)$,

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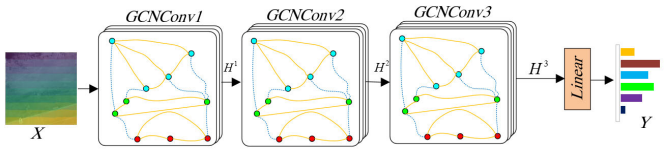


Fig. 4. GCN network architecture. it shows the GCN used in the framework, which consists of three graph convolutional layers and one linear layer. The points in the figure represent the extracted topological features, and on the right side are the classification labels for the images.

where the graph is an undirected graph. $\boldsymbol{V}$ represents node information, that is pixels, and $\boldsymbol{E}$ represents edges among nodes. These cluster center points can constitute a batch of graph data. We use M cluster centers as nodes, and the features of each node $\boldsymbol{V_i}$ are set to 5-D, $\boldsymbol{V_i} = \{r, g, b, x, y\}$, where r, g, b is the pixel value of the point, x, y is the coordinate value of the point in the original image, the advantage of this is that the position information of the pixel can be modeled, and the utilization rate of the topological structure information can be improved. The following experiments will prove it:

$$s_c = \sqrt{(r_m - r_k)^2 + (g_m - g_k)^2 + (b_m - b_k)^2}. \tag{9}$$

Then the features of these nodes form an $M \times 5$ dimensional feature matrix $\boldsymbol{X}$, and then the relationship among each node will also generate an $M \times M$ dimensional adjacency matrix $\boldsymbol{A}$. Our construction of the adjacency matrix, that is, the construction of the edge, is calculated in 2-D. The first is to use the pixel value of the center point to calculate the color difference similarity with each other. The color difference similarity equation of the two nodes $\boldsymbol{V_k}$ and $\boldsymbol{V_m}$ is as (9).

Traverse all nodes, calculate the similarity matrix among nodes $\boldsymbol{S_{color}} \in \mathbb{R}^{M \times M}$, and then calculate the mean of the matrix $\boldsymbol{S_{color}}$ as the threshold $\boldsymbol{mean_{color}}$, and use the following equation to generate the adjacency matrix $\boldsymbol{A_{color}}$:

$$A_{\text{color}} = 1, \quad if \ S_{\text{color}} > \text{mean}_{\text{color}} \text{ else } 0. \tag{10}$$

The second is to use the cosine similarity between node pixel values to generate an adjacency matrix. The cosine similarity of the two nodes $\boldsymbol{V_i}$ and $\boldsymbol{V_j}$ refers to the following equation:

$$s_\theta = \frac{V_i V_j}{\|V_i\| \times \|V_j\|} = \frac{r_i r_j + g_i g_j + b_i b_j}{\sqrt{r_i^2 + g_i^2 + b_i^2} \times \sqrt{r_j^2 + g_j^2 + b_j^2}}. \tag{11}$$

Traverse all nodes, calculate the similarity matrix among nodes $\boldsymbol{S_\theta} \in \mathbb{R}^{M \times M}$, and then calculate the mean of the matrix $\boldsymbol{S_\theta}$ as the threshold mean_θ, and use the following equation to generate the adjacency matrix $\boldsymbol{A_\theta}$:

$$A_\theta = 1 \ if \ S_\theta > \text{mean}_\theta \text{ else } 0. \tag{12}$$

We take the intersection of $\boldsymbol{A_{color}}$ and $\boldsymbol{A_\theta}$ as the adjacency matrix of $\boldsymbol{A}$. The advantage of this is that it can further reduce the number of edges among nodes and reduce the calculation amount of GCN. The experiments in the following text also prove this point:

$$A = A_{\text{color}} \cap A_\theta. \tag{13}$$

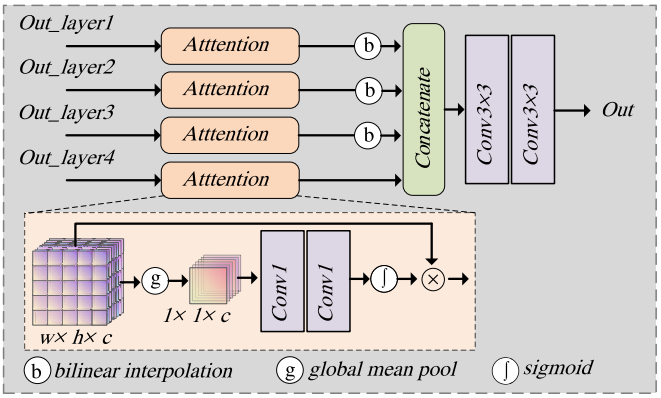


Fig. 5. MLAF structure. the input of the multilevel feature attention module is formed by concatenating the outputs of each convolutional layer in the convolutional network after dimension matching through attention mechanism. Finally, after two convolutional operations, the global fusion feature of the multilevel features is obtained.

After $\boldsymbol{A}$ is generated, we input $\boldsymbol{A}$ into GCN, and the structure of GCN is shown in Fig. 4. We uses a three-layer GCN, and the adjacency matrix $\boldsymbol{A}$ and the node feature $\boldsymbol{X}$ is input into the network, and the graph convolution process refers to the following equations:

$$H^1 = f(X, A) = \delta\left(AXW^1\right) \tag{14}$$

$$H^2 = f\left(H^1, A\right) = \delta\left(AH^1W^2\right) \tag{15}$$

$$Y = \text{linear}\left(H^3\right) = \text{linear}\left(f\left(H^2, A\right)\right) = \text{linear}\left(AH^2W^3\right)\right) \tag{16}$$

where $\boldsymbol{H^i}$ represents the input feature of the ith layer graph convolution, $f()$ represents the graph convolution operation, δ represents the activation function, $\boldsymbol{A}$ is the adjacency matrix, and W^i is the weight of the ith layer graph convolution, linear represents a linear operation, which reduces the dimensionality of the $\boldsymbol{H^3}$ features to match the number of classes.

D. MLAF Module

Due to the limitation of the size of the convolution kernel and the limited receptive field of the convolutional network, the extraction of global semantic information is insufficient. In order to make up for this defect, we design an MLAF module, as shown in Fig. 5. The input of this module is the output feature of each convolution block in the encoding stage, the purpose of this processing is to fuse feature maps of multiple scales, improve the ability to recognize objects of different scales and shapes, reduce noise and interference in the image, and improve the robustness and stability of the algorithm. In addition, we use the attention mechanism to focus on important global features to make up for the insufficient extraction of global contextual information by convolution operations. This is crucial to the overall performance improvement of the model, and the experiments in Section IV-D also show the effectiveness of this module.

The input of the MLAF module is the feature map output by the four convolutional blocks in the encoding stage, and then through the attention mechanism, taking **out_layer1** as an

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example, the specific process refers to the following equations:

$$z_c = g(\mathbf{out_layer1}) = \frac{1}{W \times H} \sum_{i=1}^{W} \sum_{j=1}^{H} \mathbf{out_layer1}(i, j) \tag{17}$$

where $\boldsymbol{g}$ represents the global average pooling operation, $\mathbf{out_layer1} \in \mathbb{R}^{H \times W \times C}$, $\boldsymbol{z_c} \in \mathbb{R}^{1 \times 1 \times C}$. Then z_c is input into the two-layer convolutional network and activated using the activation function, as shown in the following equation:

$$\alpha = \delta(F_2(F_1(z_c))) = \delta(W_2(W_1 z_c + d_1) + d_2) \tag{18}$$

where δ represents the sigmoid activation function, F_1 and F_2 represent the convolution operation, respectively, W_1 and W_2 are the weight parameters of F_1 and F_2, respectively, and d_1 and d_2 are the biases of F_1 and F_2, respectively, $\boldsymbol{\alpha} \in \mathbb{R}^{1 \times 1 \times C}$. Finally, multiply out_layer1 and the attention weight $\boldsymbol{\alpha}$ to get the output of the attention module, refer to the following equation:

$$\mathrm{Att}_1 = \mathrm{out_layer1} \otimes \alpha \tag{19}$$

where $\mathbf{Att_1} \in \mathbb{R}^{W \times H \times C}$, Since the feature size of each layer is inconsistent, after calculating the attention mechanism, we use the size of $\mathbf{Att_4}$ as the benchmark, use the bilinear interpolation function to reduce the dimension, process the size of $\mathbf{Att_1}$, $\mathbf{Att_2}$ and $\mathbf{Att_3}$ into the size of $\mathbf{Att_4}$, and then these four features are concatenated, finally after two 3×3 convolutions, the $\boldsymbol{Out}$ is obtained.

E. Objective Function

The overall training loss of MTC_Net consists of three parts: 1) topological classification loss of GCN; 2) classification loss of CNN; and 3) classification loss of the combination of the above two.

In the topological classification loss of GCN, we utilize the KL divergence loss function, which can compare the closeness between the prediction result and the true label. Refer to the following equation:

$$\begin{aligned} \mathrm{loss}_1 &= D_K L(P_G || g) \\ &= E\left[\log P_G(x) - \log(g(x))\right] = \sum_{i=1}^{N} P_G(x_i) \cdot \log \frac{P_G(x_i)}{g(x_i)} \end{aligned} \tag{20}$$

where P_G represents the output prediction value of GCN, g represents the true label, and N is the number of neurons.

When designing the classification loss function of CNN, we adopt the most commonly used cross-entropy loss function, as shown in the following equation:

$$\begin{aligned} \mathrm{loss}_2 = -\frac{1}{N} \sum_{i=1}^{N} & g(x_i) \log P_C(x_i) \\ & + (1 - g(x_i)) \log(1 - P_C(x_i)) \end{aligned} \tag{21}$$

where P_C represents the output prediction value of CNN.

Since we fuse the intermediate features of the GCN in the encoding stage of the convolutional network, we further fuse the prediction value P_G of GCN and the prediction value P_C of CNN to form a joint prediction value p and use the cross-entropy loss function calculate loss_3, as shown in the following equations:

$$p = \frac{P_G + P_C}{2} \tag{22}$$

$$\mathrm{loss}_3 = -\frac{1}{N} \sum_{i=1}^{N} g(x_i) \log p(x_i) + (1 - g(x_i)) \log(1 - p(x_i)). \tag{23}$$

Finally, the loss function of MTC_Net can be calculated by the following equation:

$$\mathrm{loss} = \lambda_1 \mathrm{loss}_1 + \lambda_2 \mathrm{loss}_2 + \lambda_3 \mathrm{loss}_3 \tag{24}$$

where λ_1, λ_2, and λ_3 are the regularization factors of loss_1, loss_2 and loss_3, respectively.

TABLE I
DATASETS INFORMATION

Task	datasets	Image_size	classes	Number	Input_size	Resolution(m)
classification	SIRI-WHU	200×200	12	2400	256×256	3
	WHU-RS19	600×600	19	1005	256×256	0.5
segmentation	Potsdam	512×512	5	5472	512×512	8
	Vaihingen	512×512	6	742	512×512	9

IV. EXPERIMENT

A. Dataset Information

We validate MTC_Net using four publicly available datasets, two classification datasets are SIRI-WHU [32] and WHU-RS19 [33], and two segmentation datasets are Potsdam [34] and Vaihingen [35], and the detailed introduction of each dataset is shown in Table I.

B. Experimental Settings

In the classification task, the size of the input image is 256×256, the batch size is 10, the initial learning rate is 0.001, the learning rate adopts the attenuation strategy with epoch, the epoch is 400, the SGD optimizer is used, the momentum is 0.9, and the weight decay is 5×10^{-5}. The number M of hyperparameter cluster centers is 256, and the regularization factors λ_1, λ_2, and λ_3 of the objective function are 0.2, 0.2, and 0.6, respectively.

In the segmentation task, the size of the input image is 512×512, the batch size is 4, the initial learning rate is 0.01, the learning rate adopts the attenuation strategy with epoch, the epoch is 400, the SGD optimizer is used, the momentum is 0.9, and the weight decay is 10^{-4}, the number M of hyperparameter clustering centers is 512. In the segmentation task, an end-to-end encoding-decoding U-Net approach is adopted. The encoding stage is the part after removing the linear operations in the classification framework, while the decoding stage is the mirror of the convolutional network in the classification framework, converting downsampling to upsampling. The cross-entropy loss used is loss_1.

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TABLE II
PERFORMANCE COMPARISON BETWEEN MGA AND SLIC ALGORITHM

Algorithm	WHU-RS19		SIRI-WHU	
	Acc	mPA	Acc	mPA
VGG	83.65	80.67	96.25	85.42
VGG+SLIC [36]	90.54	88.09	97.03	86.97
VGG+MGA	**90.97**	**88.50**	**97.27**	**87.15**

C. Evaluation Metrics

In order to evaluate the effectiveness of this method for semantic segmentation of remote sensing images, the performance analysis is performed by making a confusion matrix to calculate mPA, accuracy, F-measure ($F1$) score, Kappa coefficient, and mean intersection and union ratio(mIoU). where TP, FP, FN, and TN are true positive, false positive, false negative, and true negative classifications, respectively,

$$\begin{aligned} Accuracy &= \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{FN}+\mathrm{TN}+\mathrm{FP}} \\ \mathrm{Precision} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \\ \mathrm{Recall} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \\ F-\mathrm{measure} &= \frac{2\times \mathrm{Precision}\times \mathrm{Recall}}{\mathrm{Precision}+\mathrm{Recall}} \\ \mathrm{mIoU} &= \frac{1}{k+1}\sum_{i=0}^{k}\frac{\mathrm{TP}}{\mathrm{FN}+\mathrm{FP}+\mathrm{TP}} \end{aligned} \tag{25}$$

D. Ablation Experiments

1) Effectiveness of MGA: When obtaining the topological relationship of the image, the MGA algorithm is first used to classify adjacent pixels into the same type of pixel, represented by the pixel value of the cluster center. At the same time, other pixels are replaced by the cluster center category to generate a mask, concatenating the mask and image to form a new input feature. When using the MGA algorithm to generate the mask, there is a hyperparameter M, and M represents the number of cluster centers.

We have compared our proposed MGA method with the SLIC algorithm. As shown in Table II, We found that overall, our MGA method achieves better results compared to SLIC [36], although the improvement is not significant. However, it is worth noting that the SLIC algorithm performs better in terms of the mPA metric on the SIRI-WHU dataset. Nevertheless, our MGA method outperforms SLIC in other metrics on both datasets.

As shown in Table III, on the WHU-RS19 dataset, the final Acc and mPA of the VGG network are 83.65% and 80.67%, respectively. With the incorporation of the MGA algorithm, which concatenates the mask and the original image and input them into the network, there is a significant improvement in performance. We set the value of M, and as M increase, both Acc and mPA gradually increase. When M is set to 256, the Acc and mPA reach their highest values at 90.97% and 88.50%, respectively, which are 7.32% and 7.83% higher than VGG, respectively. When M is set to 512, Acc and mPA decrease by 1.56% and 0.95%, respectively. On the SIRI-WHU dataset, the Acc and mPA of the VGG network are 96.25% and 85.42%, respectively, while with M set to 256, the Acc and mPA are 97.27% and 87.15%, respectively, representing an improvement of 1.02% and 4.73%, respectively, which are the highest values observed for M. The same trend is observed on both the SIRI-WHU and WHU-RS19 datasets, so we believe that the value of M is related to the size of the image, and that the best performance is achieved when M is consistent with the size of the image.

TABLE III
COMPARISON OF THE RESULTS OF DIFFERENT M VALUES AND VGG RESULTS ON THE WHU-RS19 AND SIRI-WHU DATASETS

Dataset	Metrics	VGG	5	10	19	50	196	256	512
WHU-RS19	Acc	83.65	85.39	84.37	84.37	84.88	87.92	**90.97**	88.41
	mPA	80.67	83.92	82.50	83.33	83.50	87.00	**88.50**	87.15
SIRI-WHU	Acc	96.25	96.46	96.52	96.46	96.46	96.85	**97.27**	96.53
	mPA	85.42	85.76	86.81	86.28	85.68	87.08	**87.15**	86.69

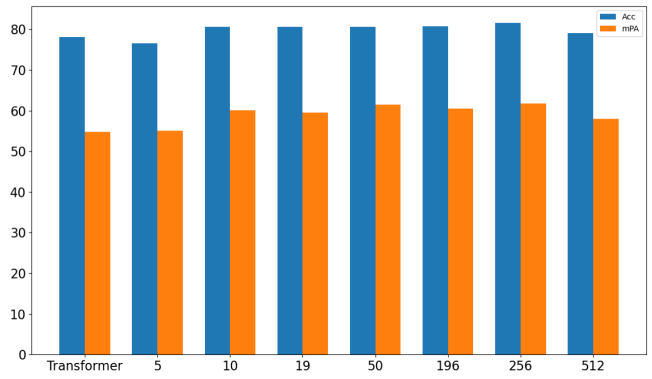


Fig. 6. Comparison of Transformer as a baseline and the variation of M value.

In order to further verify the effectiveness of the MGA algorithm, we conduct an M-value experiment on the WHU-RS19 dataset with Transformer [37] as the baseline, as shown in Fig. 6. It can be seen that when there is a mask, the performance of the Transformer is further improved. When the M value is 256, both Acc and mPA have reached the highest.

In addition, we conduct experiments on the WHU-RS19 dataset using DenseNet [38] as a baseline, and the results are shown in Table IV. The Acc and mPA of DenseNet are 87.24% and 69.67%, respectively. When we set M to 256 and concatenate the generated mask with the original image, then input it into DenseNet, the Acc, and mPA improve to 91.84% and 76.25%, respectively, representing an improvement of 4.6% and 6.58%, respectively. In summary, Table III shows that the mask is very useful for feature extraction. By conducting ablation experiments with Transformer, VGG, and DenseNet as baselines and adding the mask, the performance of the models is further improved. Through numerous ablation experiments, we have demonstrated that the performance is best when the number of clustering points is close to or equal to the size of the image, which also confirms the effectiveness of the MGA algorithm.

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TABLE IV

ON THE WHU-RS1 DATASET, THE COMPARISON BETWEEN THE RESULT OF M VALUE 256 AND THE RESULT OF DENSENET

Metrics	DenseNet	M=256
Acc	87.24	**91.84**
mPA	69.67	**76.25**

TABLE V

RESULTS OF DIFFERENT CONSTRUCTION METHODS OF ADJACENCY MATRIX

	VGG	mask	cos_similarity	dist_similarity	position	WHU-RS19		SIRI-WHU	
						Acc	mPA	Acc	mPA
1	√	X	X	X	X	83.65	80.67	96.25	85.42
2	√	√	X	X	X	90.97	88.50	97.27	87.15
3	√	√	√	X	X	93.24	90.65	97.46	87.23
4	√	√	X	√	X	93.12	90.61	97.32	87.20
5	√	√	√	√	X	93.22	90.69	97.48	87.21
6	√	√	√	√	√	**94.65**	**92.93**	**97.55**	**87.46**

2) Effectiveness of Image Topology Construction: We construct the topological structure of the image based on the generated mask, and use GCN to further extract the topological structure information of the image. The key is to convert the topological structure of the image into a graph structure, and then input it into the GCN to further extract the topological structure information of the image. In the process of establishing the graph structure, the construction of the adjacency matrix among nodes is the most important. We use three methods based on cosine similarity, chromatic distance and fusion of the two to construct the adjacency matrix, as shown in Table V.

In Table V, we show the experimental results of different adjacency matrix construction methods respectively. Comparing rows 3 with 2, both Acc and mPA are improved on both datasets. In particular, in the WHU-RS19 dataset, the improvement is particularly obvious, with Acc and mPA increasing by 2.27% and 2.15%, respectively. This shows that using GCN to extract topological structure information is effective, and it also shows that the feature fusion method is reasonable.

Comparing rows 3 and 4, both Acc and mPA have decreased on the WHU-RS19 dataset and the SIRI-WHU dataset, but the magnitude of the decrease is particularly small. It also shows that using cosine similarity to build an adjacency matrix is better than the way based on chromatic distance. row 5 is the experimental result of constructing an adjacency matrix based on the fusion of cosine similarity and chromatic distance. It can be seen that on the WHU-RS19 dataset, the Acc and mPA are 93.22% and 90.69%, respectively, which is lower than the result based solely on cosine similarity, but higher than that based solely on chromatic distance. On the SIRI-WHU dataset, this method achieves higher performance than the previous two methods. The differences in the results among the three methods are not significant, but we ultimately choose the fusion of both methods to construct the adjacency matrix. The reason is that the fusion of both methods can further simplify the adjacency relationships among nodes, and it can reduce the time for extracting topological information from the entire graph neural network, as shown in Table VI.

TABLE VI

TIME OF DIFFERENT CONSTRUCTION METHODS OF ADJACENCY MATRIX

	cos_similarity	dist_similarity	all
Make edge time	0.324ms	0.331ms	**0.280ms**

TABLE VII

EFFECTIVENESS OF MLAF

Model	WHU-RS19		SIRI-WHU	
	Acc	mPA	Acc	mPA
VGG+GCN	94.65	92.93	97.55	87.46
VGG with MALF+GCN	**96.94**	**95.50**	**98.23**	**88.69**

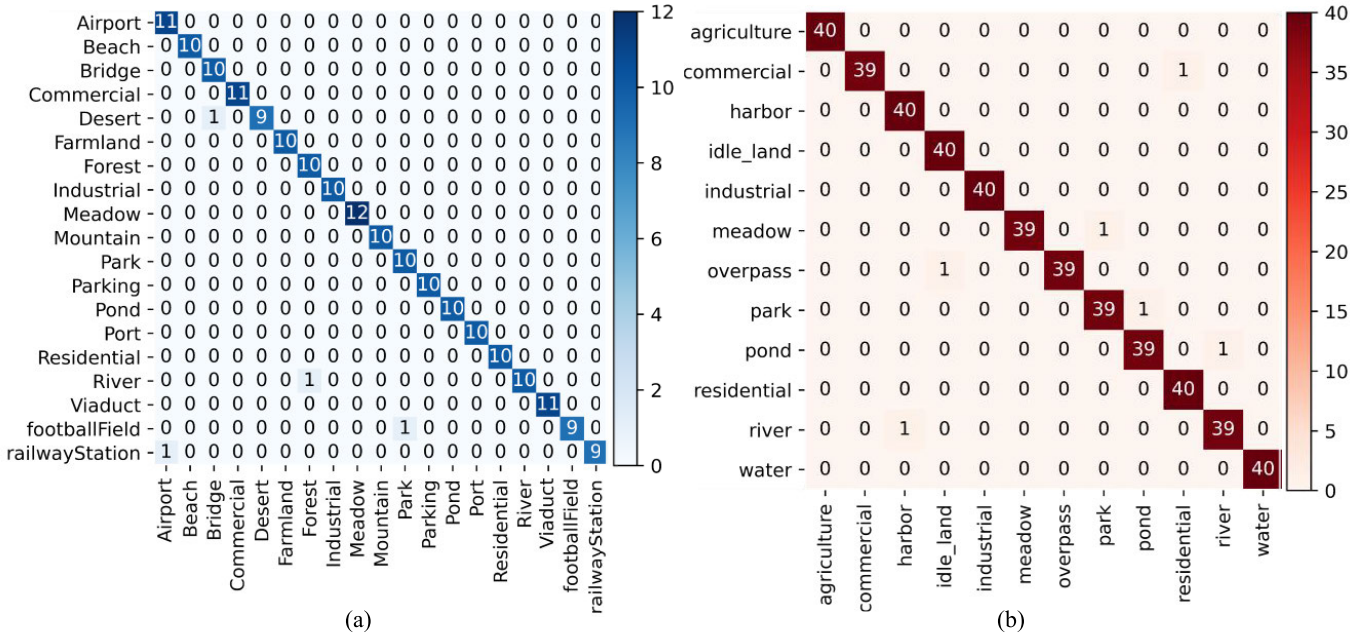
The construction time of the edges using the fusion method is the shortest among the three different methods, while the performance is relatively good. Therefore, we use the fusion method to construct the adjacency matrix.

In addition, in the process of constructing node features, we not only use the pixel information of nodes but also input the position information of nodes into GCN as a potential feature. Comparing rows 5 and 6 of Table V, we can find that, after adding location information, in the WHU-RS19 dataset, Acc and mPA increase by 1.43% and 2.24%, respectively. In the SIRI-WHU dataset, both Acc and mPA are also improved, which shows that we can improve the effect by inputting location information as feature information into GCN. In summary, by using the image topological structure information constructed from clustering centroids, and then inferencing through graph neural network, we achieve further improvement in performance, which also confirms the effectiveness of our method.

3) Effectiveness of MLAF: In order to alleviate the defect of insufficient global information extraction caused by the small receptive field of the convolutional network, we proposes a MLAF module to extract global important features of the image. Through the analysis of the above experiments, the final results of MTC_Net are shown in the Table VII. On the WHU-RS19 dataset, Acc and mPA reach 96.94% and 95.50%, respectively, which are 13.29% and 14.83% higher than VGG respectively and 2.29% and 1.57% higher than VGG+GCN, respectively. On the SIRI-WHU dataset, Acc and mPA reach 98.23% and 88.23%, respectively, which are 1.98% and 3.27% higher than VGG, and 0.68% and 1.23% higher than VGG+GCN. respectively. At the same time, we visualize the classification confusion matrix of the WHU-RS19 dataset and the SIRI-WHU dataset, as shown in Fig. 7. In summary, the MLAF can effectively extract the global information of the image to improve the performance of the overall model.

4) Analysis of Hyperparameters λ_1, λ_2, and λ_3: We have considered different combinations ofλ_1, λ_2, and λ_3, namely $[0.1, 0.1, 0.8]$, $[0.2, 0.2, 0.6]$, $[0.3, 0.3, 0.4]$, and $[0.33, 0.33, 0.33]$. The experimental results, as shown in Table VIII. The purpose of these weights, which sum up to 1, is to balance the impact of different loss functions and optimize the training process of the model, enabling it to adapt more effectively to different tasks and data distributions.

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(a) (b)

Fig. 7. Classification confusion matrix of MTC_Net on WHU-RS19 dataset and SIRI-WHU dataset.

TABLE VIII
ANALYSIS OF HYPERPARAMETERS λ_1, λ_2, AND λ_3

$[\lambda_1, \lambda_2, \lambda_3]$	WHU-RS19		SIRI-WHU	
	Acc	mPA	Acc	mPA
$[0.1, 0.1, 0.8]$	96.23	94.47	97.53	88.03
$[0.2, 0.2, 0.6]$	**96.94**	**95.50**	98.23	**88.69**
$[0.3, 03, 0.4]$	96.79	95.36	98.00	88.63
$[0.33, 0.33, 0.33]$	96.80	95.42	**98.24**	88.64

TABLE IX
COMPARISON RESULTS OF MTC_NET AND OTHER STATE-OF-THE-ART METHODS

Model	WHU-RS19		SIRI-WHU		Params(M)	FLOPs(G)
	Acc	mPA	Acc	mPA		
VGG [31]	83.65	80.67	96.25	85.42	165.77	20.30
ViT [37]	88.06	74.83	96.25	84.84	85.65	16.86
DenseNet [38]	87.24	69.67	88.75	75.63	6.966	3.783
AlexNet [39]	86.16	83.28	93.75	83.56	18.01	0.408
ResNet50 [40]	89.60	86.34	88.89	79.15	23.53	5.397
EfficientNet B0 [41]	85.62	83.08	96.04	85.87	**4.023**	**0.537**
MobileNet [42]	90.82	87.35	92.52	81.36	4.945	1.887
MLPMixer [43]	89.65	86.57	92.50	81.29	65.63	4.843
Swin [44]	81.63	78.49	93.79	82.15	27.50	4.371
PVT [45]	88.57	85.19	96.91	85.29	55.35	10.74
ConvNext [46]	91.63	88.55	94.16	84.04	87.52	20.05
Zhang et al. [47]	92.30	90.03	88.88	76.05	68.35	15.00
Yuan et al. [48]	92.37	90.11	90.26	78.54	96.27	19.72
SF-MSFormer [49]	95.58	93.86	97.24	88.41	36.31	21.41
ours	**96.94**	**95.50**	**98.23**	**88.69**	70.63	9.221

E. Comparison With State-of-the-Art Methods

1) Classification: To further validate the effectiveness of MTC_Net, we compare it with several representative methods, as shown in Table IX. MTC_Net considers how to capture the topological features of the image and the semantic features of objects with different scales and shapes. In contrast, the existing state-of-the-art methods do not take these two points into consideration. The advantage of MTC_Net lies in its design specifically addressing these two aspects, which is the reason why our method achieves better results.

On the WHU-RS19 dataset, MTC_Net achieve Acc of 96.94% and mPA of 95.50%, which is an improvement of 13.29% and 14.83%, respectively, compared to VGG [31]. The Acc and mPA of ResNet50 [40] are 89.6% and 86.34%, respectively, which are 7.34% and 9.16% lower than those of MTC_Net, respectively. We also compare MTC_Net with recent popular methods based on the Transformer mechanism. The Acc and mPA of Swin [44] are 81.63% and 78.49%, respectively, which are improved by more than 10% by MTC_Net, and MTC_Net improves by 5.31% and 6.95% over ConVNexT [46], respectively. In addition, MTC_Net exhibits better performance compared to the latest methods. The Acc and mPA of the method in [47] are 92.30% and 90.03%, respectively, which are 4.64% and 5.47% lower than those of MTC_Net, respectively. Compared with the method in [48], MTC_Net improves by 4.57% and 5.39%, respectively. Although the latest SF-MSFormer [49] method also achieve excellent results, with Acc of 95.58% and mPA of 93.86%, it still do not surpass the results of MTC_Net.

On the SIRI-WHU dataset, MTC_Net still demonstrates strong advantages, outperforming both representative classic methods and recent methods based on the Transformer mechanism, as well as the latest methods. The Acc and mPA of MTC_Net are 98.23% and 88.69%, respectively, while the Acc and mPA of EfficientNet B0 [41] are 96.04% and 85.87%, respectively. It is clear that MTC_Net is superior.

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TABLE X
SEGMENTATION EFFECTS OF VARIOUS GROUND OBJECTS ON THE POTSDAM SEGMENTATION DATASET

Classes	Acc	F1	Kappa	mIoU
Impervious surfaces	95.03	94.72	88.79	88.34
Building	**97.32**	96.52	**89.67**	89.98
Tree	87.59	87.75	86.38	86.51
Car	97.12	**96.93**	89.42	**91.05**
Low vegetation	89.64	88.38	85.34	85.72
Overall	93.34	92.86	87.92	88.32

The Transformer-based PVT [44] method performs worse than MTC_Net, with a decrease of 1.32% and 3.4%, respectively. Compared with the latest methods [47], [48], [49], MTC_Net achieves the best results in terms of Acc and mPA.

In addition, we compare MTC_Net with other methods in terms of the number of parameters and floating-point operations (FLOPs). As shown in Table VII, the Params and FLOPs of MTC_Net are 70.63M and 9.221G, respectively, which are at a moderate level compared to the methods that we compare with. In summary, MTC_Net demonstrates the best performance on both datasets, and its Params and FLOPs are lower than that of most methods, which further illustrates that MTC_Net's computational complexity and resource requirements can be run on common platforms. This indicates that MTC_Net is sensitive and efficient for the classification of remote sensing images.

2) Segmentation: In order to further verify the effectiveness of the MTC_Net, we also verify it on two public segmentation datasets. The experimental results show that MTC_Net can also perform excellent results in the segmentation task of remote sensing images.

On the Potsdam dataset, we calculate the Acc, $F1$, Kappa, and mIOU for various ground objects, which are 93.34%, 92.86%, 87.92%, and 88.32%, respectively, as shown in Table X. MTC_Net performs the worst in the segmentation of low vegetation. Through our observations of the images, we find that the distribution of low vegetation is uneven, and there are significant differences in low vegetation between different locations, which lead to inaccurate segmentation by MTC_Net. However, MTC_Net demonstrates significant advantages in segmenting small objects, with a segmentation accuracy of 97.12% for cars. In the meantime, we also visualize some of the test results on the Potsdam dataset, as shown in Fig. 8. It can be seen that the segmentation boundary of low vegetation is inaccurate, and there are many misclassified edge pixels, but the segmentation of cars is ideal. The segmentation of other ground objects is also very efficient.

We also compare MTC_Net with other advanced methods, as shown in Table XI. "-" indicates that the metric is not reported in the corresponding paper. It can be seen that MTC_Net outperforms other methods on all metrics. Compared with GLSANet [16], MTC_Net improves by 2.93% and 5.39% in Acc and mIoU, respectively. The classic method

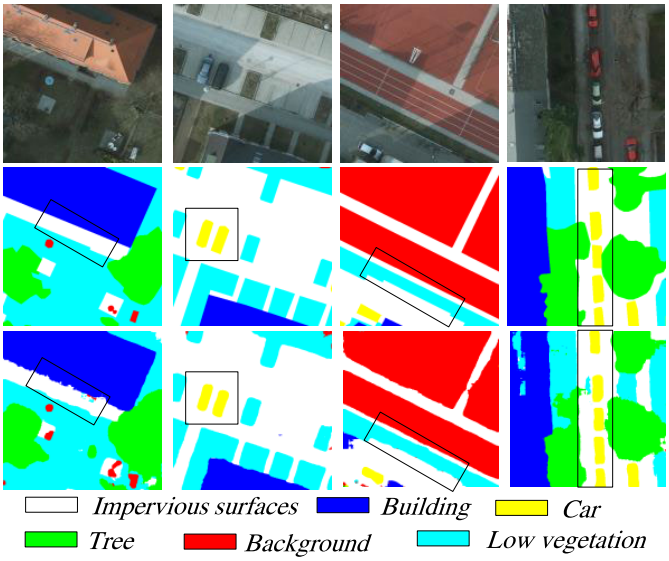


Fig. 8. MTC_Net visualizes some images on the Potsdam dataset. The first row is the original image, the second row is the corresponding GT, and the third row is the prediction result of MTC_Net.

TABLE XI
COMPARISON WITH OTHER ADVANCED METHODS ON THE POTSDAM SEGMENTATION DATASET

Method	Acc	F1	Kappa	mIou
GLSANet [16]	90.41	-	-	82.93
DeepLabV3+ [50]	91.09	87.55	-	-
DC_Swin [51]	92.00	**93.25**	-	87.56
BIBED-Seg [52]	90.80	-	**89.80**	76.00
SBSSNet [53]	-	-	-	87.68
MISNet [54]	84.73	-	-	75.64
Jin et al. [55]	84.60	-	-	75.36
Guan et al [56]	91.99	87.95	-	-
MCAFNet [57]	90.20	89.80	-	88.20
Xiao et al. [58]	88.12	83.59	-	73.62
ours	**93.34**	92.86	87.92	**88.32**

DeepLabV3+ [50] achieves Acc of 91.09% and $F1$ of 87.55%, which are still lower than those of MTC_Net. BIBED-Seg [52] achieves a Kappa of 89.80%, which is 1.88% higher than that of MTC_Net, but MTC_Net outperforms BIBED-Seg by 2.54% and 12.32% in Acc and mIoU, respectively. Compared with the latest method [58], MTC_Net improves Acc, $F1$ and mIoU by 5.22%, 9.27%, and 14.7%, respectively. This improvement is very huge. The metrics of dc_Swin [51] are the closest to MTC_Net, but It is still lower than MTC_Net in terms of Acc and mIoU. This also shows the advancement of the method.

In order to further prove the effectiveness of MTC_Net, we also conduct experiments on the Vaihingen dataset. The segmentation information of various ground objects is shown in Table XII. On the Vaihingen dataset, MTC_Net still shows excellent performance. The overall Acc, $F1$, Kappa and mIOU

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TABLE XII
SEGMENTATION EFFECTS OF VARIOUS GROUND OBJECTS ON THE VAIHINGEN SEGMENTATION DATASET

Classes	Acc	F1	Kappa	mIou
Impervious surfaces	96.33	92.92	88.89	87.84
Building	**97.62**	94.58	89.27	89.98
Tree	86.89	86.95	85.78	86.51
Car	97.02	**95.13**	**90.12**	**90.75**
Low vegetation	90.34	88.18	86.34	82.72
Overall	93.64	91.55	88.08	87.56

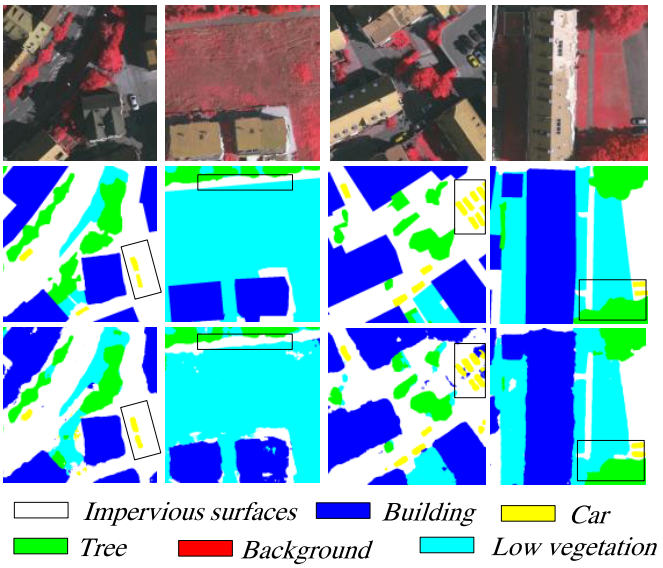


Fig. 9. Visualization of some images of MTC_Net on the Vaihingen dataset. The first row is the original image, the second row is the corresponding GT, and the third row is the prediction result of MTC_Net.

are 93.64%, 92.86%, 86.34%, and 87.56%, respectively, but the segmentation accuracy of tree is the worst. The good news is that MTC_Net has a very good segmentation effect on small targets such as car. The visualization of some testset is shown in Fig. 9, which basically shows a similar segmentation rule with the Potsdam dataset. The segmentation of car is very accurate, but the classification of the boundary pixels of tree is relatively vague, and the classification is not particularly accurate.

At the same time, we compare the MTC_Net with other advanced methods, as shown in Table XIII, "-" in the table means that this metrics does not appear in the corresponding whole paper. It can be seen that the MTC_Net is optimal in all metrics. The Acc and $F1$ of the classic method DeepLabV3+ [50] are 91.93% and 90.71%, respectively, but still lower than MTC_Net. Compared with MISNet [54], MTC_Net improves Acc and mIoU by 4.01% and 5.97%, respectively, and the Kappa of BIBED-Seg [52] is 83.70%, which is 4.38% lower than MTC_Net. Compared with the latest method [62], MTC_Net improves Acc, $F1$ and mIoU by 5.13%, 6.38%, and 13.44%, respectively. This improvement is very huge. The metrics of Long et al. [60] are the closest to MTC_Net, but

TABLE XIII
COMPARISON WITH OTHER STATE-OF-THE-ART METHODS ON THE VAIHINGEN SEGMENTATION DATASET

Method	Acc	F1	Kappa	mIou
DeepLabV3+ [50]	91.93	89.39	-	-
DC_Swin [51]	91.63	90.71	-	83.22
BIBED-Seg [52]	87.70	-	83.70	69.60
MISNet [54]	89.63	-	-	81.59
Jin et al. [55]	89.48	-	-	81.15
Guan et al [56]	88.42		-	**87.95**
Xiao et al. [58]	86.97	81.38	-	69.79
Deng et al. [59]	92.05	90.54	-	-
Long et al. [60]	91.93	91.45		84.45
RSSFormer [61]	90.84	90.57	-	-
He et al. [62]	88.51	85.17		74.12
ours	**93.64**	**91.55**	**88.08**	87.56

It is still lower than MTC_Net in the three metrics of Acc, $F1$, and mIoU. In summary, MTC_Net has also shown strong capabilities in segmentation tasks, which further illustrates the advancement of MTC_Net.

V. CONCLUSION

In this article, we propose a novel framework called MTC_Net that can extract topological information with global context for complex image categories, addressing the limitations of convolutional neural networks in capturing nonlocal topological information. MTC_Net efficiently extracts complex features from remote sensing images, providing valuable insights for complex image feature extraction. Extensive experiments on four public datasets demonstrate its excellent performance in classification and segmentation tasks.

We find that fusing the topological structure information with semantic information is highly effective. In future work, our key research focus will be on accurately representing the topological structure information of images and improving the model feature extraction capability for complex base objects. Although our method achieves good results, the parameter size of the model is moderate. We will concentrate on designing a lightweight model and balancing various metrics. Our goal is to develop a unified framework that can cater to multiple tasks.

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